

Charlie Marx

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EDUCATION

Stanford University
Ph.D. in Computer Science
Advisor: Stefano Ermon

2020 - PRESENT

Haverford College
B.S. in Computer Science, *highest honors*
B.S. in Mathematics, *highest honors*
GPA: 3.96/4.0, *magna cum laude*

2016 - 2020

AWARDS & HONORS

NSF Graduate Research Fellowship
Stanford School of Engineering Fellowship
Phi Beta Kappa
CRA Outstanding Undergraduate Researcher Runner-Up
Arnold and Mabel Beckman Foundation Award
Farmers Insurance Standout Student of the Year

2021
2020
2020
2019
2018
2016

PUBLICATIONS

MACHINE LEARNING METHODS

1. [Modular Conformal Calibration](#)
Charles T. Marx*, Shengjia Zhao*, Willie Neiswanger, Stefano Ermon
ICML – *International Conference on Machine Learning*, 2022
2. [Predictive Multiplicity in Classification](#)
Charles T. Marx, Flavio du Pin Calmon, Berk Ustun
ICML – *International Conference on Machine Learning*, 2020
Previously Selected for Oral Presentation at 2019 NeurIPS HCML Workshop
3. [Disentangling Influence: Using Disentangled Representations to Audit Model Predictions](#)
Charles T. Marx, Richard L. Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian
NeurIPS – *Conference on Neural Information Processing Systems*, 2019

MACHINE LEARNING APPLICATIONS

4. [Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill](#)
Helen White, **Charles T. Marx**, David Valentine, Charles Sharpless, Christoph Aeppli, Kelsey Gosselin, Veronika Kivenson, Rachel Liu, Robert Nelson, Sean Sylva, Christopher Reddy
ACS Earth and Space Chemistry, 2019
5. [Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence](#)
Anna Michel, Alexandra E. Morrison, **Charles T. Marx**, Helen White
ACS Environmental Science & Technology Letters, 2018

RESEARCH EXPERIENCE

Research Intern *Amazon Web Services*

SUMMER 2022

Designed methods for quantifying uncertainty in model explanations.
Advisors: Youngsuk Park and Hilaf Hasson

Summer Intern *International Food Policy Research Institute*

SUMMER 2020

Developed machine learning tools to classify land usage in Africa using satellite images.
Advisors: Channing Arndt and Zhe Guo

Research Assistant in Machine Learning *Harvard University*

SUMMER 2019

Studied predictive multiplicity: the ability for datasets to admit multiple equally accurate models with conflicting predictions. Developed integer programming tools and software to measure the severity of multiplicity. Compiled findings into first-author conference and workshop paper.
Advisors: Flavio Calmon and Berk Ustun

	Summer Research Intern <i>Prince William Sound Science Center</i> Automated salmon and herring tracking in Alaska using machine learning with sonar data. <i>Advisor: Scott Pegau</i>	SUMMER 2018
ACADEMIC ACTIVITIES	Teaching Assistant in Machine Learning Held weekly office hours and graded for an upper-level machine learning course.	2019 – 2020
	Peer Tutor Tutored students in statistics, linear algebra, and machine learning.	2018 – 2019
	Computer Science Faculty Search Committee Student representative on the computer science tenure-track faculty search committee.	2018 – 2019
INVITED & CONTRIBUTED TALKS	Simons Institute for the Theory of Computing <i>Berkeley, CA</i> Disentangling Influence (45-minute talk)	2019
	INFORMS Annual Meeting <i>Seattle, WA</i> Predictive Multiplicity in Classification (20-minute talk)	2019
	FAT* Conference <i>New York, NY</i> Auditing Black Box Models (60-minute tutorial) <i>with Carlos Scheidegger and Suresh Venkatasubramanian</i>	2018
	GOMOSSES Conference <i>New Orleans, LA</i> Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15-minute talk)	2018
POSTER PRESENTATIONS	ICML Conference <i>Baltimore, MD</i> Modular Conformal Calibration	2022
	ICML Conference <i>Vienna, Austria</i> (virtual) Predictive Multiplicity in Classification	2020
	NeurIPS Conference <i>Vancouver, Canada</i> Disentangling Influence: Using Disentangled Representations to Audit Model Predictions	2019
	NeurIPS Workshop on Human-Centric Machine Learning <i>Vancouver, Canada</i> On the Multiplicity of Predictions in Machine Learning	2019
	Beckman National Symposium <i>Irvine, CA</i> Indirect Influence in Machine Learning Models	2019
	Beckman Regional Symposium <i>New York, NY</i> Mitigating Social Discrimination in Deep Learning	2018
	KINSC Undergraduate Research Symposium <i>Haverford, PA</i> Examining Biogenic Hydrocarbons After the Deepwater Horizon Oil Spill	2017
FIELDWORK	Alaska Prince William Sound Research Charter	2018
	Gulf of Mexico Shoreline Research Tour	2017
ACADEMIC SERVICE	Reviewing: KDD '20, ICML '22 Program Committee: NeurIPS '20 Workshop on Fair AI in Finance, FAccT '22	
EXTRA– CURRICULAR ACTIVITIES	Captain of Haverford College Men's Club Soccer	2018 – 2019
	Leader of Haverford College Outdoors Club	2017 – 2018
	President of Furniture for Newly Housed Families Organization	2015 – 2017
SKILLS & INTERESTS	Languages: <i>Fluent:</i> Python, R, LaTeX · <i>Familiar:</i> SQL, C, Unix Tools: PyTorch, Tensorflow, Pandas, CPLEX Interests: Backpacking, Gardening, Basketball, Folk Music, Cooking	